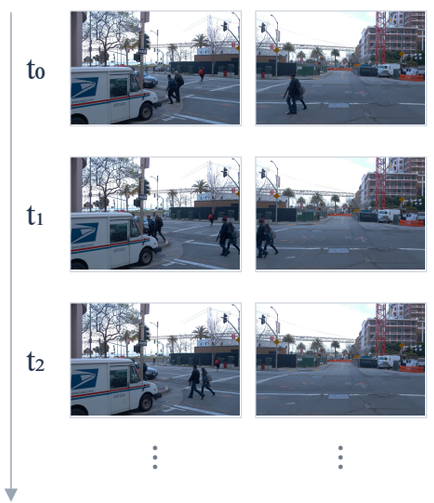


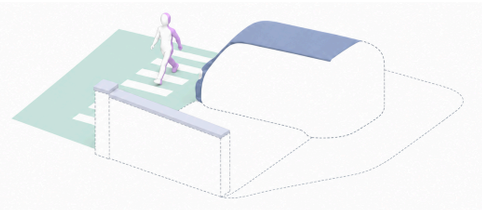
(a) Previous methods: state changes expose missing surfaces

Observation-only reconstruction



Camera change

State-driven rendering



(b) Ours: joint completion enables asset reuse across states

Shared scene context





